



UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT



Data Structure and Algorithm
Laboratory Activity No. 4

Arrays

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I. Objectives

Introduction

Array, in general, refers to an orderly arrangement of data elements. Array is a type of data structure that stores data elements in adjacent locations. Array is considered as linear data structure that stores elements of same data types. Hence, it is also called as a linear homogenous data structure.

This laboratory activity aims to implement the principles and techniques in:

- Writing algorithms using Array data structure
- Solve programming problems using dynamic memory allocation, arrays and pointers

II. Methods

Jenna's Grocery

Jenna's Grocery List		
Apple	PHP 10	x7
Banana	PHP 10	x8
Broccoli	PHP 60	x12
Lettuce	PHP 50	x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in

Jenna's Grocery List.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.

III. Results

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Grocery List  
Apple - ₦ 10 × 7 = ₦ 70  
Banana - ₦ 10 × 8 = ₦ 80  
Broccoli - ₦ 60 × 12 = ₦ 720  
Lettuce - ₦ 50 × 10 = ₦ 500  
  
Total Cost: 1370  
  
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Removing Lettuce  
-----  
  
Updated Grocery List  
Apple - ₦ 10 × 7 = ₦ 70  
Banana - ₦ 10 × 8 = ₦ 80  
Broccoli - ₦ 60 × 12 = ₦ 720  
-----
```

Figure 1 Screenshot of output of the program

The program successfully created two subclasses, "Fruit" and "Vegetable", that inherited from a base class "GroceryItem". Each item was stored as an object inside the array GroceryList. The program computed the total sum using a function that iterated through all elements and calculated the sum of prices multiplied by quantities.

IV. Conclusion

This laboratory activity helped me understand how arrays work and how they can be used to store and organize data in a program. Using Python lists as arrays made it easy to add, display, and remove items. I also learned how to create classes for different objects like fruits and vegetables and how to calculate their total prices.

References

- [1] Co Arthur O.. "University of Caloocan City Computer Engineering Department Honor Code," UCC-CpE Departmental Policies, 2020.